Westline TPMS: methodology and limitations

How the work was done

Data used. The five client documents (brief, data dictionary, fleet manager transcript, task sheet, inflation spec), the data inventory, and four daily summary tables produced from the combined TyreSense file (559,179 rows, 1 June to 15 July 2026, six trucks, 14 fitted slots each; axle 6 and 7 slots are empty by design and excluded).

Not analysed. The vehicle master, tyre register, tread-depth sheets and daily kilometre log were seen only as an inventory, not analysed. Findings that need them are marked open in the findings file.

Tools. ChatGPT wrote and helped run the two profiling scripts and drafted the first package. Claude analysed the summary tables, wrote the findings, deck, distrust memo and checking script (analysis_checks.py). Both transcripts are in the AI Transcript folder.

Reasoning order

1. Physics first: the spec standard is a cold pressure, and gas pressure follows absolute temperature, so running readings cannot be compared with it directly. 2. Cold baseline: idle Sundays (sensor temperature about 28 C) approximate cold pressure; each truck was compared with its axle standard. 3. Cross-check against the manager: his depot routines (cold inflation at Belagavi, hot inflation at Hubballi) were tested against the size and shape of the shortfall, and his leak explanation was tested against drift between fills. 4. Dual pairs: the full daily inner/outer gap series was read, not only the largest values. 5. The July failure was read against the paired tyre and the battery level.

Known limitations

Approximate statistics. The event script combines chunk-level results by averaging medians and shares and taking the minimum of chunk 10th percentiles. These are not exact daily statistics. They were used only to compare trucks and to spot steps, never as exact values. Exact values need recomputing from raw rows.

The candidate file is not used for conclusions. candidate_low_pressure_days.csv flags 72.6% of all slot-days because it compares running pressure with a cold target, so it does not separate faults from normal operation.

Vehicle IDs. KA25AB4471 and KA25B4471 are treated as one truck. Duplicate rows were not removed, because only daily tables were available.

Sunday baseline. It assumes trucks stood idle and sensor temperature was near ambient. One truck (KA48AF9054) reads 0.997 on 5 July and may have run that day.

Unconfirmed. Which trucks belong to which depot, the timing of fills, the rotation date and the sensor-to-slot mapping after it, and the meaning of inspection codes C and D (the dictionary and the manager conflict).

What would change the conclusions

Hourly raw data showing fills happening in the morning on the low trucks, a downward drift between fills, or sensor temperatures far above ambient on Sundays would weaken the hot-inflation explanation. A decline visible on 10 July in KA25AC1182 raw readings would change the lead-time conclusion for the July failure.

Validation plan

Recompute daily metrics from raw rows with exact quantiles; confirm the vehicle-sensor-tyre map with effective dates; align telemetry to register and repair records; test alert rules on a time-ordered holdout; report lead time, false alerts per truck-week and missed events.